



Forest Appeals Commission

Fourth Floor, 747 Fort Street
Victoria BC V8W 3E9
Telephone: (250) 387-3464
Facsimile: (250) 356-9923

Mailing Address:
PO Box 9425 Stn Prov Govt
Victoria BC V8W 9V1

Website: www.fac.gov.bc.ca
Email: facinfo@gov.bc.ca

DECISION NO. 2017-WFA-003(a)

In the matter of an appeal under section 40(1) of the *Wildfire Act*, S.B.C. 2004, c. 31.

BETWEEN:	Andersen Pacific Forest Products Ltd.	APPELLANT
AND:	Government of British Columbia	RESPONDENT
BEFORE:	Jeffrey A. Hand, Panel Chair	
DATE:	June 11 – 20, 2018	
PLACE:	Victoria, BC	
APPEARING:	For the Appellant: Mark S. Oulton, Counsel Nicole C. Gilewicz, Counsel For the Respondent: Johnny Van Camp, Counsel	

APPEAL

[1] This appeal arises out of harvesting activities undertaken by Andersen Pacific Forest Products Ltd. (“Andersen”) on Timber Sale Licence A82206 (the “Licence”), Block C059C3HT, known as Parkinson Creek, just south of Port Renfrew on Vancouver Island. Andersen undertook two fire hazard abatement programs at Parkinson Creek following the completion of harvesting operations. However, on June 30, 2017, Ken Taekema, Fire Centre Manager with the Coastal Fire Centre, Ministry of Forests, Lands and Natural Resource Operations (the “Ministry”)¹, issued Fire Hazard Abatement Order No. 2017-CoFC0002 (the “Abatement Order”), requiring Andersen to undertake further fire hazard abatement at Parkinson Creek. The Order was issued under section 7(3) of the *Wildfire Act* (the “Act”).

[2] Andersen appeals the Abatement Order on the grounds that the risk of fire ignition and spread at Parkinson Creek is very low due to the wet site and the local weather, and that no further abatement is required. Accordingly, the Abatement Order ought to be rescinded. In the alternative, Andersen argued during its closing submissions that the order is overly broad and ought to be varied.

¹ As of July 18, 2017, the Ministry became the Ministry of Forests, Lands, Natural Resource Operations and Rural Development.

[3] The Government of British Columbia (the “Government”) asserts that the Abatement Order is necessary to address an unacceptable fire risk, and asks that the Abatement Order be confirmed and the appeal dismissed.

[4] The Forest Appeals Commission has the power to hear this appeal pursuant to section 39(1) of the *Act*. The Commission’s powers on an appeal are set out in section 41(1) of the *Act*, as follows:

41(1) On an appeal under section 39 by a person ... the commission may

- (a) consider the findings of the decision maker who made the order, and
- (b) either
 - (i) confirm, vary or rescind the order, or
 - (ii) with or without directions, refer the matter back to the decision maker who made the order, for reconsideration.

RELEVANT LEGISLATION

[5] The following provisions are relevant to this appeal:

The Act

Hazard assessment and abatement

7(1) ...

- (2) A person, other than a person who is in a prescribed class of persons referred to in subsection (2.1), carrying out an industrial activity or a prescribed activity, must, within a prescribed time period and to the prescribed extent, abate a fire hazard of which the person is aware or ought reasonably to be aware.

...

- (3) Despite subsections (2) and (2.1), if an official identifies circumstances that the official considers constitute a fire hazard in relation to

- (a) an industrial activity, or

- (b) a prescribed activity referred to in subsections (2) and (2.1),

the official by written order may require the applicable person to abate the fire hazard by a specified date. [Emphasis added]

...

[6] There is no dispute that Andersen’s harvesting operations were an “industrial activity” as defined in the *Act*.

Wildfire Regulation, B.C. Reg. 38 / 2005**Hazard abatement: qualified holders****12.1 (1) ...**

- (3) A person required to abate a fire hazard under section 7 (2) of the Act who is a qualified holder must reduce the fuel hazard on the site of the industrial activity or prescribed activity
- (a) as necessary to ensure that carrying out the activity
 - (i) does not increase the risk of a fire starting on the site, and
 - (ii) if a fire were to start, would not increase the fire behaviour or fire suppression associated with the fire, or
 - (b) as specified by a professional forester or registered forest technologist under the *Foresters Act*.

CHRONOLOGY AND SUMMARY OF THE EVIDENCE***The Licence and Harvesting at Parkinson Creek***

[7] Andersen was the successful bidder for the Licence under the BC Timber Sales ("BCTS") Program. The Licence was issued to Andersen on March 28, 2014, allowing Andersen to undertake harvesting activities at Parkinson Creek.

[8] Parkinson Creek is located on the west side of Vancouver Island, approximately four kilometres south of Port Renfrew. The northeast boundary of the block is adjacent to Highway 14. There are electric power lines running along Highway 14 that supply power to Port Renfrew.

[9] Parkinson Creek is located approximately two kilometres inland from the Pacific Ocean. It has a southwestern exposure with a slope varying between 0 to 30%, with an average slope of approximately 15%. It is located at an elevation of approximately 250 metres.

[10] The tree species within the block consists of Cedar, Hemlock and Douglas Fir, but it is dominated by Cedar.

[11] Parkinson Creek is located in a very wet maritime variant of a biogeoclimatic ecosystem ("BEC") zone, classified as Coastal Western Hemlock variant CWHvm1. The CWHvm1 is characterized by significant precipitation and humidity, mild winters, and a long growing season.

[12] BCTS prepared a Project Risk Rating Form for Parkinson Creek after the Licence was issued to Andersen. This form is used by BCTS to assess the frequency of any inspections they intend to conduct during the harvesting operations. The Project Risk Rating Form also includes BCTS's assessment of the "Fire Hazard and Ease of Ignition" at Parkinson Creek. In this instance, it was scored by BCTS as 1, which is "low risk". Rod Clark of BCTS testified that he believed that the fire hazard was higher than a score of 1, but he could not say why the score on the form did not reflect this view.

[13] Andersen commenced harvesting in June, 2014.

[14] Angus Hope, P.Eng., Registered Professional Forester ("RPF"), was Andersen's manager for these harvesting operations. He testified that he was on site from time to time in the summer of 2014. He said that, on those days, weather conditions varied. Some days were sunny, some were rainy and foggy, and some were cloudy. No weather records were kept by Andersen of the actual weather conditions on site during Andersen's operations at Parkinson Creek.

[15] Mr. Hope testified that he is very familiar with the area around Parkinson Creek. He said that the area experiences frequent fog, even in the summer months, and that mornings are often cool. He considered Parkinson Creek to be in an area called the "fog zone", which extends along portions of the west coast of Vancouver Island.

[16] On August 20, 2014, Mr. Clark of BCTS performed an inspection at Parkinson Creek. Falling and yarding operations were complete and loading of timber was just beginning. His inspection comments included the following:

There is heavy slash along the hiway [sic]. The Licensee should consult with the Wildfire management branch on the requirements for fire Hazzard assessment and treatment along this Hiway [sic].

[17] Roy Vidler of BCTS next inspected Parkinson Creek on September 3, 2014. His report includes this comment: "Avoid creating slash piles close to the highway".

[18] Andersen completed its harvesting and loading operations at Parkinson Creek in approximately late November or early December, 2014. Thereafter, Andersen obtained a waste survey to determine the volume of material left on site. That survey estimated that the material remaining on site was at a concentration of 49 tonnes per hectare.

Andersen's Pile Burning Programs

[19] In April, 2015, Andersen carried out a pile burning program at Parkinson Creek. Andersen created piles of the remaining harvesting debris left on site after all marketable timber had been removed. This consists of mostly smaller debris, referred to throughout these proceedings as "slash", along with other larger pieces of waste timber. Material smaller than seven centimetres in diameter is referred to as "fines". The piles are ignited in what is expected to be a controlled burn. The intent of the burn program is to consume most of the fine fuel remaining on site, such that the block is left in a condition that does not increase the risk of fire beyond that which existed prior to harvesting operations. It is also to ensure that the ability to suppress a fire, or the ability of a fire to spread, has not been increased beyond what existed prior to harvesting operations.

[20] Andersen retained Kelly McKinnon, RPF, to prepare a burn plan for the first burn performed in April, 2015. Mr. McKinnon testified that he designated only some of the piles to be burned, choosing to leave others unburned to aid in the regeneration of the site. According to both Mr. Hope and Mr. McKinnon, this burn was largely unsuccessful because the piles did not burn sufficiently. Mr. Hope testified that this was due to the local wet conditions associated with the CWHvm1

BEC zone, while other witnesses testified that the weather conditions generally in April are not conducive to undertaking successful burn programs. The Panel will return to that evidence later in these Reasons.

[21] Andersen chose to undertake a second pile burning program in late November, 2015. Mr. McKinnon prepared a pile burning map, which designated 26 piles to be burned. These piles were located predominantly along the sides of the in-block roads used by Andersen during the harvesting process. A particularly large pile located immediately adjacent to Highway 14 was not burned because of concerns that smoke from that pile could create safety issues along the highway due to reduced visibility.

[22] Following the second burn program undertaken by Andersen in November, 2015, Parkinson Creek came under the scrutiny of both BCTS and the BC Wildfire Service.

[23] Mr. Clark testified that, as part of his typical follow-up on BCTS licences, he reviewed Parkinson Creek to ensure that the Licence requirements for silviculture, deactivation, and fire hazard assessment were completed so that the security deposit that Andersen had paid to BCTS in respect of the Licence could be returned.

[24] Representatives from BC Wildfire Service had also observed the conditions at Parkinson Creek in or about the end of 2015 while performing routine follow-up. Alan Berry, Registered Forest Technologist ("RFT"), was the Zone Supervisor and Forest Protection Officer with the BC Wildfire Service. He advised Mr. Clark in a January 11, 2016 email that:

There were a couple of roadside accumulations I would have preferred they included in their fall burning but they did look to have burned most of the piles within the block. The bigger concern was the dispersed slash which was well over threshold levels and considering the proximity to the highway, may need further abatement. Can you forward me a contact number for the licensee so I can request a FHA [fire hazard assessment]

[25] Mr. Berry contacted Mr. Hope on January 21, 2016, and requested a fire hazard assessment for Parkinson Creek. When no fire hazard assessment was immediately forthcoming, Mr. Clark contacted Mr. Hope again on March 29, 2016, to enquire as to the status of completion of the fire hazard assessment, as well as requesting that Andersen complete the deactivation and grass seeding activities at Parkinson Creek. Deactivation encompasses returning any drainage courses at Parkinson Creek that had been disturbed during harvest activities to their original condition, as well as taking measures to prevent public access to the in-block roads.

Andersen's Fire Hazard Assessment

[26] Andersen engaged Mr. McKinnon to conduct a post-harvest fire hazard assessment. Mr. McKinnon did so in or about March, 2016, utilizing a guide issued by the Wildfire Management Branch in April, 2012, known as "A Guide to Fuel Hazard Assessment and Abatement in British Columbia" (the "Guide").

[27] The introduction to the Guide is instructive and relevant to this appeal. It states:

This Guide has been created to provide a default procedure for determination of the fire hazard created by the carrying out of an industrial or prescribed activity on forest land and the degree of abatement needed to reduce fire hazard if the hazard has been increased by the activity. The objective of legislation is to ensure that fire hazards are identified and abated in an appropriate time and manner and to a level that ensures fuel hazards do not increase the potential fire behaviour and suppression effort associated with a fire start. It is particularly important to ensure that fires do not threaten communities, critical infrastructure or other significant values. (page 6)

[28] The Introduction then goes on to state:

.... It is important to note that given the variation in conditions across British Columbia it [the Guide] cannot properly represent all areas, thus modifications may be warranted to reflect regional conditions and specific localized circumstances. (page 6)

[29] Section 2 of the Guide sets out a three-step process to determine whether a fire hazard needs to be abated. The first step is to "Determine the Location of the Potential Fire Hazard".

[30] Mr. McKinnon testified that he determined the location of the potential fire hazard by reviewing the British Columbia Fire Risk Map. This map designates areas across the Province on a scale containing four classifications: low, moderate, high and severe. Those areas of severe and high rating are typically located in and around urban areas due to the risk to infrastructure and public safety associated with a fire starting close to those urban areas. Parkinson Creek is designated as a severe risk on this map owing to its proximity to Port Renfrew.

[31] However, Mr. McKinnon testified that he considered the risk of fire ignition at Parkinson Creek to be low, rather than severe, because of the localized climatic conditions associated with the CWHvm1 BEC zone. In doing so, relied on section 2 of the Guide which provides:

... local topographic features and their modifying effect on fire spread and direction may also be considered as factors that reduce potential wildfire impacts. Other considerations for non abatement of fuels may include no or highly limited access and/or low ignition probability due to constant wet conditions (e.g. coastal hyper maritime and other very wet areas). (page 12)

[32] It should be noted that Mr. McKinnon was not called as an expert witness to express an opinion on the need for further abatement at Parkinson Creek. Rather, he was called as a factual witness to provide evidence of the steps taken by Andersen to assess the fire risk at Parkinson Creek. Whether, and to what extent, the localized conditions at Parkinson Creek impact the need for abatement was

addressed by expert witnesses called by both parties, and will be discussed later in these Reasons.

[33] Step 2 in the Guide is to “Determine the Fuel Hazard Rating”. It determines the fuel hazard present on a given site by considering the dominant woody species, the tonnes per hectare of debris, the slope, and the aspect of the block.

[34] With minor exceptions, all of the experts that considered the condition at Parkinson Creek, including Mr. McKinnon, concluded that the dominant fuel type on site was classified in the “Canadian Forest Fire Danger Rating System” as “S-3”, being made up mostly of Cedar, with a density of 20 to 49 tonnes per hectare. There is also little disagreement that the slope of Parkinson Creek was, on average, 15%, and that its aspect was generally southwest.

[35] With these characteristics in mind, Appendix 1 of the Guide sets out a fuel hazard chart which assigns a numerical value to S-3 fuel type of this density on a cutblock with the slope, and aspect, of Parkinson Creek. This results in a value of approximately 70 to 80. This range is the result of averaging the Guide’s metrics for slope and aspect. Mr. McKinnon used a value of 73.

[36] Step 3 in the Guide’s assessment process is to take the numerical value determined in Step 2 (e.g., 73), and compare it to the threshold for abatement based on the risk class found on the Fire Risk Map discussed in Step 1 of the Guide. The threshold abatement chart found in the Guide is reproduced as follows:

<u>Risk Map Class</u>	<u>Fuel Hazard Threshold</u>
Low	100
Moderate	70
High	40
Severe	25

[37] The Guide recommends that abatement be performed when the fuel hazard threshold meets or exceeds the threshold set out in the chart above.

[38] Mr. McKinnon testified that, since he concluded that the localized weather conditions at Parkinson Creek presented a low risk of fire ignition, the fuel hazard rating of 73 was within the threshold of 100 for low risk. He recommended to Andersen that no further abatement was necessary. Had Mr. McKinnon found under the Guide that the risk of ignition at Parkinson Creek was moderate, high, or severe, the fuel hazard calculation that he performed would likely have resulted in a recommendation for further abatement.

[39] On April 13, 2016, Mr. Hope (Andersen) provided Mr. Clark (BCTS) with the requested fire hazard assessment prepared by Mr. McKinnon. Mr. Clark, in turn, forwarded the assessment to Mr. Berry (BC Wildfire Service) on April 14, 2016.

Events leading up to the Abatement Order

[40] On April 18, 2016, Mr. Berry advised Mr. Clark in an email that he had concerns about the volume of material and roadside accumulations that had been left on Parkinson Creek in close proximity to the highway and power lines, and that he wished to raise these issues with Mr. Hope in due course. Mr. Berry indicated that, with the approaching fire season, he would not be able to speak to Mr. Hope in the short term.

[41] In the meantime, Mr. Clark completed his final inspection report for BCTS on April 19, 2016, confirming that road deactivation had been completed and that natural drainage patterns had been restored. Notably, the report also indicated that the fire hazard assessment and abatement had been conducted as required. Mr. Clark arranged for the release of Andersen's deposit, being the final step in the completion of the Licence as far as BCTS was concerned.

[42] On May 13, 2016, Mr. Clark notified Mr. Hope concerning this final inspection in these terms:

... That is the end of BCTS obligations. Protection [a reference to Wildfire Services Branch] was concerned about the FHA [fire hazard assessment] not taking into account the slash along the hiway [sic]. That is their jurisdiction and they work under the wildfire act so you may hear from them in the future but that does not affect your deposit refund by BCTS.

[43] In September, 2016, after the summer fire season had subsided, Mr. Berry asked one of his technicians, Daniel Taudin-Chabot, an experienced fighter of wildfires but without fire hazard assessment experience, to undertake a fire hazard assessment of Parkinson Creek. Mr. Taudin-Chabot was not qualified as an expert to express an opinion in these proceedings. His evidence was confined to the observations that he made at Parkinson Creek, along with the photographs that he took on site.

[44] On December 7, 2016, Mr. Berry contacted Mr. McKinnon and provided him with the fire hazard assessment prepared by Mr. Taudin-Chabot, along with the photographs of the site. Mr. Berry asked Mr. McKinnon and Mr. Hope to meet with members of BC Wildfire Service on site to discuss the possible need for further abatement at Parkinson Creek.

[45] Thereafter, efforts were made to conduct a site visit on February 9, 2017, but this visit was postponed due to adverse weather conditions. The parties considered meeting later in the month of February but not all participants were available. Consideration was then given to meeting on site on March 13, 2017.

[46] The March meeting was cancelled at Mr. McKinnon's request, as he believed that much of Parkinson Creek was still covered in snow at that time. As it turned out, the parties were never able to meet at Parkinson Creek.

[47] On April 12, 2017, Mr. Berry requested that Morgan Boghean, RFT, a Forest Protection Technician with BC Wildfire Service, perform a fire hazard assessment on Parkinson Creek. Mr. Boghean was qualified to give expert evidence in this appeal.

Mr. Boghean concluded that further abatement at Parkinson Creek was required to remove more of the debris remaining on site, principally in the areas alongside the roadways.

[48] On May 30, 2017, following receipt of Mr. Boghean's report, Mr. Berry advised Mr. McKinnon that he continued to have concerns regarding the volume of post-harvest slash and roadside accumulations at Parkinson Creek, which caused him to believe that further abatement was required. A meeting was arranged with Mr. Hope and Mr. McKinnon to take place on June 29, 2017 at the Errington Fire Base.

[49] The Errington meeting proceeded, although Mr. Berry was unable to attend. In his absence, Clint Parker attended along with Mr. Boghean. Mr. Parker was a Senior Wildfire Officer – Prevention, with the BC Wildfire Service. Mr. McKinnon and Mr. Hope attended on behalf of Andersen.

[50] Mr. Parker states that he raised concerns about unburnt piles and dispersed slash in the area between the in-block roads and Highway 14, and expressed his view that further abatement should occur in these areas. Mr. McKinnon and Mr. Hope expressed their view that the risk of fire ignition was low, and that no further abatement was required. Ultimately, the parties were unable to resolve their differing views on whether the remaining piles and dispersed slash on site presented a fire hazard and should be further abated.

[51] The following day, Mr. Parker recommended that Ken Taekema, the Fire Centre Manager, issue a hazard abatement order requiring Andersen to perform further abatement at Parkinson Creek.

[52] On June 30, 2017, Mr. Taekema issued the Abatement Order. The order states, in part:

Pursuant to section 7(3) of the *Wildfire Act*, I have identified circumstances that I consider constitute a fire hazard in relation to timber harvesting operations that occurred on TSL A82206 Block C059C3HT Parkinson Creek.

The circumstances constituting the fire hazard are as follows:

Unabated post-harvest roadside accumulations (piles) and dispersed fuels fire hazard associated with timber harvesting operations.

As authorized under section 7(3) of the Act, I order you to abate the above-noted fire hazard by *December 1, 2017*. [Italics in original]

[53] On July 25, 2017, Andersen filed this appeal. Under section 36(1) of the Act, the Abatement Order is stayed pending resolution of the appeal.

[54] As of the date of the Abatement Order, Andersen presented evidence that the Licence has been replanted, the Licence area has undergone a number of growing seasons and has experienced significant natural regeneration and seedling growth.

The Parties' Positions on the Appeal

[55] Andersen submits that a purposive interpretation of the legislation in its proper context supports a finding that any fire hazard in relation to industrial activity carried out on the Licence was fully and adequately abated, and that no abatement order is warranted under the *Act*. It argues that the Abatement Order is incorrectly premised on fire hazard assessments that: (a) fail to properly consider or give full weight to the impact that local climatic conditions have, and will continue to have, on the risk of fire ignition and spread in the Licence area; (b) fail to properly consider or give full weight to the effect of the two burn programs already undertaken by Andersen; (c) fail to properly consider or give full weight to the effect of the time and growing seasons that have passed; and (d) contain errors and misunderstandings of the legislative obligations imposed upon industry in their appropriate context. Further, it submits that no practical purpose is served by further abatement at this point in time.

[56] Andersen placed particular emphasis on the low risk of fire ignition and spread being due to the location of Parkinson Creek in the "fog zone" on the west coast, one of the wettest parts of the Province.

[57] Andersen submits that the Government issued the Abatement Order without advising Mr. McKinnon of its concerns, and despite the BCTS expressly advising Andersen that it was in compliance with its obligations under its Licence and returning Andersen's deposit.

[58] In closing, Andersen states:

The very wet nature of the local climate is full answer to any suggestion that further abatement is warranted. Simply put, there is minimal risk of fire ignition or spread in the Licence area due to the pervasive wetness of the site. The limited number and size of historical fires in this area further evidences this reality and this reality is not affected by the values at risk.

[59] The Government notes that this is the first appeal from an order issued under section 7 of the *Act*. It submits that the fact that a forestry professional retained by Andersen concluded that no further abatement is required at the site is no answer to the Abatement Order issued by an official and requiring further abatement. Moreover, the Government submits that Andersen's forestry professional was wrong. The Government further submits that the return of the deposit is irrelevant, and did not affect Andersen's continuing hazard abatement obligations under the legislation.

[60] The Government submits that Andersen incorrectly states that there is minimal or very low risk of fire starting on the block or spreading because it is in the fog zone. It submits that Andersen places too much emphasis on a single factor in the risk assessment: i.e., the fact that the block is within the CWHvm1 BEC zone.

[61] The Government further refers to both fire history in the area of the block, as well as BC's recent fire history, as being relevant factors to consider. Regarding the latter, the Government states that BC's 2017 wildfire season was unprecedented in

terms of the amount of land burned, costs of fire suppression, and people evacuated. It notes that “a Provincial State of Emergency was declared on July 7th and lasted 70 days until September 15, 2017, which was the longest in BC’s history and the first declared since 2003.”

[62] For these reasons and more, the Government submits that Andersen has not met its burden of proving that the Abatement Order was unreasonable. It submits that the appeal ought to be dismissed and that Andersen should abate the remaining fire hazards to the satisfaction of the BC Wildfire Service. It submits that the Fire Centre Manager exercised his delegated, statutory authority reasonably and in accordance with the *Act*.

ISSUES

[63] The main issue to be decided is whether further fire hazard abatement is required on the Licence area in the circumstances? Specifically, under section 7(3) of the *Act*, are there circumstances in Parkinson Creek that constitute a fire hazard, such that Andersen ought to be required to abate the fire hazard by a specified date?

[64] Prior to deciding this issue, a number of preliminary issues were raised which will be addressed before turning to the main issue. These issues are:

- What is the applicable standard of review that should be applied to the decision below in the context of this appeal? Is the Fire Centre Manager’s decision entitled to deference such that Andersen has the burden of proving that the Fire Centre Manager’s discretion was exercised unreasonably?
- Is the severity of a fire season in the Province a relevant consideration to a licensee’s abatement efforts?
- Does the fact that the concerns were identified after Andersen’s deposit was returned and BCTS had said abatement and fire assessment had been completed, operate to prevent issuance of Abatement Order?
- What is the relevance of the fog zone when assessing the fire hazard in this case?

DISCUSSION AND ANALYSIS

- 1. What is the applicable standard of review that should be applied to the decision below in the context of this appeal? Is the Fire Centre Manager’s decision entitled to deference such that Andersen has the burden of proving that the Fire Centre Manager’s discretion was exercised unreasonably?**

[65] While not strenuously argued by the Government, there was language in its submissions which suggested that the Fire Centre Manager’s decision to issue the Abatement Order was entitled to deference, and that the Commission need only determine whether his decision was reasonable.

[66] Andersen submits that no such deference is required.

[67] The Panel accepts Andersen's submissions in this regard.

[68] The Commission's powers under section 41(1)(a) of the *Act* state that, on an appeal, the Commission "may" consider the findings of the decision maker who made the order. Further, the Commission has the authority to "conduct an appeal by way of a new hearing" (sections 40.1 of the *Act* and 140.6 of the *Forest and Range Practices Act*).

[69] This appeal was conducted as a new hearing and there is evidence before the Panel that was not before the Fire Centre Manager. Moreover, this is not a judicial review. Accordingly, the Fire Centre Manager's decision is not entitled deference.

2. Is the severity of a fire season in the Province a relevant consideration to a licensee's abatement efforts?

[70] Throughout this appeal, the Government took the position that the 2017 wildfire season in the Province, being particularly severe, creates a heightened requirement for abatement efforts. In effect, abatement efforts that might have been appropriate in prior years are no longer sufficient. The Panel rejects this submission.

[71] Firstly, the Abatement Order was issued before the 2017 wildfire season. Therefore, it was not a consideration in the 2016 fire hazard assessments or the Fire Centre Manager's decision to issue that order. Further, as the 2017 wildfire season occurred in a completely different part of the Province, the Panel finds that it is not relevant new evidence for the purposes of this decision on appeal.

[72] Therefore, the Panel finds that the severity of the 2017 wildfire season is not relevant to Andersen's abatement efforts.

3. Does the fact that the concerns were identified after Andersen's deposit was returned and BCTS had said abatement and fire assessment had been completed, operate to prevent issuance of Abatement Order?

[73] The evidence reveals that Mr. Berry, on behalf of BC Wildfire Service, began having concerns about the sufficiency of the abatement efforts undertaken by Andersen by December, 2015. He raised these concerns with Mr. Clark at BCTS, but not with Andersen.

[74] Counsel for Andersen emphasizes that BC Wildfire Service did not speak to Andersen about its concerns with the level of abatement at Parkinson Creek until 2017, some eight months after delivery of Mr. McKinnon's fire hazard assessment. In the interim, BCTS had returned Andersen's deposit and noted that Andersen's obligations to perform abatement, and submit a fire hazard assessment, had been completed.

[75] The Panel finds that BCTS's decision to return Andersen's deposit is not determinative of the issues on this appeal. Mr. Clark's observation that Andersen had performed abatement and submitted its fire hazard assessment was correct –

they had done so. But he expressed no opinion on the adequacy of the abatement efforts. Indeed, he specifically advised Andersen that BC Wildfire Service might be following up with Andersen on the issue of the adequacy of its abatement efforts.

[76] The Panel also finds that nothing turns on the timing of the communications between Andersen and BC Wildfire Service. Each seems to suggest that the other is to blame for the delay. The Government points to Andersen's delay in performing the two burns at Parkinson Creek, and then to the subsequent passage of time until April, 2016 when Mr. McKinnon's fire hazard assessment was delivered.

[77] Andersen is critical of the fact that it was not told of BC Wildfire Service's concerns until 2017. However, the evidence also reveals that BCTS raised concerns over the accumulation of slash along the roadside at least twice during harvesting operations while completing their inspections. Andersen received those inspection reports, so it cannot say that it was not warned, to some extent, about concerns related to high volumes of slash.

[78] In the Panel's view, none of this evidence assists in determining the issue on this appeal, being, whether further abatement at Parkinson Creek is required.

4. What is the relevance of the fog zone when assessing the fire hazard in this case?

[79] Considerable mention was made in this appeal of an area along a portion of the west coast of Vancouver Island that the Coastal Fire Centre designated as "the fog zone". The fog zone covers an area approximately two kilometers inland from the ocean which is known to experience fog on a regular basis. Its southern boundary begins at Owen Point, which is just north of Port Renfrew, then extends northward along the west coast of Vancouver Island where it terminates just before Port Hardy.

[80] The Government's experts testified that Parkinson Creek is not found within the fog zone map. The Government submits that Parkinson Creek is not located within the fog zone since it is about four or five kilometres south of Owen Point. For this reason, the Government submits that the fog zone is irrelevant to this appeal. Further, even accepting that Parkinson Creek can experience fog as a result of its location within the CWHvm1 BEC zone, the Government maintains that further fire abatement is still required at Parkinson Creek because the risk of wildfire ignition, and the potential harm that it would pose to interface values, is unacceptably high.

[81] Andersen submits that the weather conditions experienced at Parkinson Creek are not dissimilar to those found in the fog zone. It takes the position that Parkinson Creek is, effectively, in the fog zone, regardless of whether it "officially" falls within the boundaries designated by the Coastal Fire Centre. As a result, the risk of fire ignition at Parkinson Creek should be considered "low" for the purposes of fire hazard assessment.

[82] The significance of the fog zone is that this area can be exempted from campfire bans by the Coastal Fire Manager as it generally has lower wildfire risks

than surrounding areas. A Bulletin from the Wildfire Service describes the fog zone as follows:

The fog zone was recognized as having a different climate (green). In 2009 the Coastal Fire Centre defined the area so that it could be identified on maps, could be legally defined for prohibitions, and could be better described to the public.

The idea of the fog zone was to clearly define the area so that when prohibitions were put in place this region could be excluded or included as conditions dictated. [Emphasis added]

Since this area is subject to frequent and heavy fog, it tends to be wetter and generally has lower wildfire risks than surrounding areas. This does not mean that a prohibition will not, or cannot be enacted in this area but it is an area that is often excluded from prohibition as due to damp conditions it is at lower risk for wildfire. In 2015, however, a campfire prohibition did extend to the fog zone.

[83] Little specific evidence was led by either side concerning the creation of the fog zone, or how or when campfire exemptions might be triggered. No qualified weather experts testified about the weather at Parkinson Creek. However, the Panel is satisfied that fog does not stop at Owen Point, and that areas within two kilometres of the ocean south of Port Renfrew, including Parkinson Creek, can experience fog.

[84] The Wildfire Bulletin referred to above provides some evidence of a recognition that wildfire ignition can be lower in areas that experience fog. However, this appears to be no more than a general statement, since the Bulletin also makes reference to the fog zone being either excluded or included in campfire prohibitions as conditions dictate, suggesting that wildfire ignition can still be a concern in the fog zone. Whether Parkinson Creek is inside or outside of the fog zone is not determinative of the issues in this appeal. The issue of concern is whether, having regard to all of the circumstances, there is need for further fire hazard abatement at Parkinson Creek. General statements about the presence of fog, are, in the Panel's view, not particularly helpful when those statements are unsupported by specific meteorological evidence.

5. Whether further fire hazard abatement is required on the Licence area in the circumstances? Specifically, under section 7(3) of the Act, are there circumstances in Parkinson Creek that constitute a fire hazard, such that Andersen ought to be required to abate the fire hazard by a specified date?

[85] Andersen and the Government tendered expert evidence on the issue of whether there is need for further abatement of a fire hazard risk at Parkinson Creek. The Panel will review that evidence below.

Bruce Blackwell

[86] Mr. Blackwell, RPF, holds a distinguished Forest Professional designation from the Association of BC Forest Professionals. He was called as an expert witness by the Government.

[87] Mr. Blackwell holds a Master's Degree from the University of British Columbia, Faculty of Forestry. He has studied fire ecology, which considered the effects of prescribed fire burns related to soils, vegetation, fuels and the implications to operational forestry practices. As such, he has studied specifically the process of ignition in a variety of forest fuels. Mr. Blackwell was qualified to express opinions on:

- wildfire behavior,
- fire ecology,
- wildfire management,
- fire hazard assessment, and
- fire hazard abatement.

[88] Mr. Blackwell conducted a field assessment at Parkinson Creek on December 13, 2017, during which time he walked the majority of the Parkinson Creek block. Mr. Blackwell observed what he considered to be high accumulations of fine and coarse debris alongside Highway 14 and adjacent to the in-block roads. He identified 57 piles, 25 of which were partially burned with the remainder being unburned. In his opinion, the slash materials within the partially burned piles were poorly consumed, in part, due to the poor construction of the piles.

[89] In Mr. Blackwell's opinion, a significant amount of the slash at Parkinson Creek is comprised of fine Cedar material, which is considered volatile and to have high fire behaviour. These fines, which are essentially dead limbs from fallen trees, dry out over time becoming easier to ignite, and can produce fast moving fires once they do ignite. He states that it can take 10 years for Cedar fines to completely decompose.

[90] In Mr. Blackwell's opinion, because of the proximity of Parkinson Creek to Port Renfrew, the adjacent highway, and the BC Hydro transmission line, there are high risk values to be protected which, in turn, require a high standard of abatement given the potential impact on these values should a wildfire ignite.

[91] In Mr. Blackwell's opinion, the fact that Parkinson Creek is located in the CWHvm1 BEC zone did not lower the severe risk rating on the BC Fire Risk Map to a low risk. He testified that the BEC zone designations are principally intended as a tool for species selection, rather than for fire assessment. Mr. Blackwell explains that the BEC zone classifications are based on average conditions throughout the year and are not a useful tool for assessing fire risk, which typically occurs in the months of July and August. In his view, it is the fire danger days in those months that are most determinative when assessing the risk of ignition. This is because fine material can dry out rapidly during dry warmer periods associated with the summer months, and this drying can occur in as little as 10 hours when the

conditions are right. In his opinion, even a site like Parkinson Creek that experiences wet conditions for much of the year can dry out in the summer months.

[92] Utilizing weather data from the TS San Juan weather station, Mr. Blackwell found that, during the last 11 years, Parkinson Creek experiences an average of 26 days of extreme or high fire danger risk in the summer months.

[93] Considering all of the foregoing, Mr. Blackwell was of the opinion that Andersen's two burn programs at Parkinson Creek had not abated the fire risk to an acceptable level, and that further abatement should be performed by burning the remaining piles adjacent to the in-block roads, as well as collecting heavy areas of slash into new piles.

[94] In support of his opinion, Mr. Blackwell also performed fire hazard assessments utilizing two approaches generally accepted as the basis for fire hazard assessment in British Columbia: the first involves the use of the Guide; the second uses the BCTS Strait of Georgia Fire Hazard Assessment Form. Both approaches consider the fuel hazards present on site and the risk of ignition, based on the degree to which the site is accessible to the public. The latter is considered a useful metric because the vast majority of wildfires are caused by human activity.

[95] Using the Guide, Mr. Blackwell began his analysis by noting that the classification of Parkinson Creek is in the severe fire risk category due to its proximity to Port Renfrew. Any score over 25 utilizing the Guide triggers abatement for severe fire risk locations.

[96] In Mr. Blackwell's opinion, the "S-3" debris on site had a loading that exceeded 50 tonnes per hectare, with a slope of greater than 15% and a southerly aspect. With only slight variations, these numbers are not seriously in dispute and were, more or less, used by the other experts who testified. This type of fuel in this concentration results in a fuel hazard rating of 100, well above the threshold of 25 for a severe risk category.

[97] As noted above, Mr. Blackwell did not believe that the local climatic conditions at Parkinson Creek justified an adjustment from severe fire risk to low fire risk. In cross-examination he noted that, even if he assumed a somewhat lower fire risk, either in the high or moderate category, this would still exceed the threshold that triggers abatement according to the Guide.

[98] Using the BCTS Strait of Georgia Fire Hazard Assessment Form, Mr. Blackwell generated a score of 32. This is based on his visual observations of the depth of fuel on site, its size, and its vertical and horizontal arrangement. Abatement is generally deemed appropriate for any score over 14. Thus, Mr. Blackwell's score of 32 is more than double the score of 14, the threshold for triggering the need for hazard abatement. Such a score suggests that, even allowing for some variation in the scoring values, a score in excess of 14 seems very likely.

[99] In terms of the degree to which the site is accessible to the public (a metric used by both approaches), Mr. Blackwell testified that he saw evidence of campfire residue, spent rifle shells and some garbage, all indications that the public is able to access Parkinson Creek.

[100] Mr. Blackwell stated that the area in and around Parkinson Creek has historically experienced approximately 68 wildfire since 1950. He agreed that wildfires are less common in CWHvm1 as compared to drier subzones, but he pointed out that they do occur, and that they can be large fires.

[101] In addition to expressing his own opinion about the need for further abatement at Parkinson Creek, Mr. Blackwell was also asked to comment on the fire assessment prepared by Mr. McKinnon in 2016. Mr. Blackwell was critical of Mr. McKinnon's assessment, principally for the following reasons:

- Mr. McKinnon's use of the BEC zone classification to justify lowering the risk classification from severe to low is an improper use of annual average conditions, without considering the fire danger in the summer months;
- there was insufficient consideration of the fine fuels present on site and their impact on ignition potential and fire behavior;
- there was no consideration of fire history in the area, and the role of human caused ignition;
- there was insufficient recognition given to the vulnerability of adjacent infrastructure (the power line) and to the community of Port Renfrew; and
- there was a lack of detail and supporting documents provided to underlie Mr. McKinnon's assessment.

[102] In cross-examination, Mr. Blackwell agreed that performing a fire hazard assessment is a matter for forestry professionals, and that those professionals are not required to necessarily follow either the Guide or the BCTS Strait of Georgia assessment tool: i.e., the legislation does not specify a method. Rather, a fire hazard assessment is "results-based", meaning that the objective is to assess whether abatement is necessary to reduce the fire hazard.

[103] Notwithstanding Mr. Blackwell's reluctance to consider the fire hazard at Parkinson Creek to be low because of its CWHvm1 BEC classification, he did agree that local weather is important for assessing the risk of ignition. He agreed that any weather station data used to determine fire danger days needs to be representative of the local weather experienced at a given site.

[104] Mr. Blackwell also agreed that topography can affect local weather, and that the TS San Juan weather station is located at a higher elevation than Parkinson Creek and at a distance approximately 10 kilometres inland from the ocean. He agreed that, on a given day, marine fog might be present at Parkinson Creek, but not at the TS San Juan weather station.

[105] Mr. Blackwell also agreed that some portions of Parkinson Creek have wet soils. However, he observed that much of the fine fuels that were causing him concern were not in contact with the ground. Rather, those fuels were elevated and, as such, would still burn even though the ground beneath them may remain wet.

[106] Mr. Blackwell did not perform a detailed assessment of the slash volume. Rather, based on his visual observations, he assessed the Cedar slash component as falling in the range of 40% to 60%. Mr. Blackwell said that the Cedar slash was easy to pick out because it is gray/white in colour. Indeed, many of the photographs taken at Parkinson Creek show this gray/white slash. Mr. Blackwell agreed in cross-examination that some of the piles that he observed were properly constructed, while others were not.

William Lougheed

[107] Mr. Lougheed was called as an expert by Andersen. He has approximately 45 years of experience in various roles within the forest industry, much of it in the field of assessing wildfire hazards and implementing hazard abatement programs. Mr. Lougheed has overseen in excess of 6,000 individual controlled burns. He was an active RFP until he retired in November, 2017, before seeking re-instatement with the Association in the spring of 2018. Mr. Lougheed was qualified to express opinions on fire hazard assessment and fire hazard abatement.

[108] Mr. Lougheed prepared two expert reports which were tendered into evidence in this proceeding.

[109] Mr. Lougheed's first report was dated February 17, 2018. For this report, he was asked to comment on the considerations that go into conducting a fire hazard assessment and, based on his review of the three Fire Hazard Assessments for the block by Mr. McKinnon, Mr. Taudin-Chabot and Mr. Boghean, to provide his opinion on: (a) the appropriate fire risk category for the block; and (b) whether further fire abatement is required for the block.

[110] Mr. Lougheed did not conduct a site visit at Parkinson Creek prior to preparing this report.

[111] In Mr. Lougheed's opinion, no further abatement was required at Parkinson Creek, because:

- (a) two burn programs had already been completed when conditions were right for controlled burns;
- (b) Parkinson Creek is in the fog zone;
- (c) access to the site was restricted;
- (d) any fine material on site would likely have decomposed three years post-harvest; and
- (e) the area around Parkinson Creek is not known for its abundance of wildfires.

[112] Mr. Lougheed agreed with Mr. McKinnon's opinion that Parkinson Creek had a low risk of ignition, despite having a severe risk rating as shown on the BC Fire Risk Map. In his opinion, with the passage of time, the fire risk was continuing to lessen as the site experienced "green-up" through the growth of new seedlings and grasses. Since Parkinson Creek is located in the CWHvm1 zone, Mr. Lougheed believed that decay of fine material would occur quickly and that, since two burn

programs had been completed already, it was unlikely that any significant fine material remained on site.

[113] Mr. Lougheed also noted that Parkinson Creek would be at a low fire hazard for “the bulk of the year”. He agreed that fire danger days are an important consideration when determining fire risk, and that Parkinson Creek will dry out during periods associated with what are considered extreme or high fire danger weather conditions. Mr. Lougheed reviewed the weather data from the TS San Juan weather station and calculated that extreme or high danger class days were limited to no more than 7.8 days per year on average. However, in cross-examination, Mr. Lougheed conceded that he had made a mathematical error and that the actual fire danger days averaged 26 days per year.

[114] Mr. Lougheed assumed that access to the site had been restricted so there was no concern about public access to Parkinson Creek. However, in cross-examination he agreed that, when he went to the site after the first report was prepared, he was able to easily gain access to the block and to drive his pickup truck along the block roads.

[115] Mr. Lougheed’s second report was prepared on March 13, 2018, following a March 9, 2018 site visit. On that site visit, Mr. Lougheed observed 68 piles on site. Of these, he observed that 52 had been completely burned, meaning that they were 85% to 95% consumed by fire. Of the remaining 16 piles, Mr. Lougheed believed that only four were potential candidates for further burning, because the other 12 did not have sufficient fine material to ignite. In cross-examination, Mr. Lougheed agreed that those four piles should have been abated and that, had he been overseeing the burn program, they would have been completely consumed.

[116] Mr. Lougheed performed a fire hazard assessment utilizing the BCTS Strait of Georgia Fire Hazard Assessment Form. Mr. Lougheed scored Parkinson Creek as 16, which is more than the abatement threshold of 14. Nevertheless, in his opinion, no further abatement was needed because: (a) any debris on site was rapidly degrading, (b) most of the piles, in his view, had already been completely burned, and (c), any attempt at further abatement would cause damage to the tree seedlings that had been planted following the harvesting operations.

[117] In cross-examination, Mr. Lougheed agreed that the proximity of Parkinson Creek to Highway 14, the power line, and the community of Port Renfrew, were values that needed to be considered in any hazard assessment out of a desire to protect those values from loss following a wildfire.

[118] In terms of the local climatic conditions of Parkinson Creek, Mr. Lougheed considered its classification in the CWHvm1 BEC zone as justification for moving Parkinson Creek out of the severe fire risk category to the low risk category. He explained that wetter climates typically make for more difficult ignition and, as well, wet sites typically lead to faster degradation of fine material which, in turn, lowers the fire risk further. In Mr. Lougheed’s opinion, it is the summer immediately after harvesting that presents the greatest risk of fire due to the presence of fine materials on site. However, these generally degrade over time and, as they do, they continue to reduce the risk of fire. In his second report, Mr. Lougheed said that the fine Cedar material could degrade in as few as three years, whereas, in

cross-examination he agreed that it could take as much as 10 years for the Cedar fines to fully degrade.

[119] Mr. Lougheed states that he is generally familiar with this area of Vancouver Island and, in his experience, it experiences foggy conditions throughout the year on some days. While Parkinson Creek is not, strictly speaking, within the fog zone designated by BC Wildfire Service, he said that the climate experienced at Parkinson Creek is not dissimilar to that which could be expected in the designated fog zone.

[120] Mr. Lougheed placed some reliance on the fact that two burn programs had been undertaken by Andersen and that, if unburned piles remained, this was indicative of the low ignition characteristics of Parkinson Creek due to the wet conditions. In cross-examination, he conceded that it was possible that the two unsuccessful burns could be due to poor pile construction, rather than due to a wet site. Mr. Lougheed also conceded that Andersen's attempt to undertake burns in the months of April and November/December were not the best times of year to carry out controlled burns, and that there were better months at the end of the fire season, but before the winter season, to undertake this work.

[121] Mr. Lougheed also agreed that the dominant species on site was Cedar, and that Cedar is a volatile type of slash. Mr. Lougheed did not disagree that there were some areas on the site that had heavier amounts of slash than others, even after the two burn programs performed by Andersen.

[122] Mr. Lougheed was of the opinion that there was no practical benefit to be obtained by undertaking further abatement because three growing seasons have now passed since the completion of harvesting operations, and the risk of ignition continues to lessen with each passing year. He also expressed concern that undertaking further abatement would likely damage the regeneration of seedlings and grass that has now started on site.

Morgan Boghean

[123] Mr. Boghean was called to give expert evidence for the Government. He is an RFT with the BC Wildfire Service and has experience fighting wildfires on the fire line. Mr. Boghean also has experience conducting fire cause and origin investigations and fire hazard assessments for BCTS. Mr. Boghean was qualified to give opinion evidence in relation to fire hazard assessment and fire hazard abatement, as well as wildfire management.

[124] Mr. Boghean completed a fire hazard assessment at Parkinson Creek following a site visit that he performed in April, 2017. He returned to Parkinson Creek in September, 2017, and took some photographs.

[125] During his site visits, Mr. Boghean observed evidence of multiple camp fires having taken place at Parkinson Creek. He also observed tire tracks from vehicles within the block.

[126] Mr. Boghean testified that the depth of slash present on the site varied, but was of significant depth, particularly in the area between the in-block roads and Highway 14. He did not measure the slash, but estimated the amount based on

visual observations. In his opinion, the percentage component of Cedar within the slash was in the order of 60% to 70%. Mr. Boghean estimated that the total debris loading throughout the site was greater than 50 tonnes per hectare.

[127] Mr. Boghean considered the risk of ignition to be high, principally because of the close proximity of unburned or partially burned piles to the in-block roads, which would put them in close proximity to human activity should people enter the block. In his opinion, should a fire ignite at Parkinson Creek, it would be difficult to fight because of the depth of the slash component remaining on site, and the possibility that the fire could spread quickly, and easily jump over Highway 14.

[128] Mr. Boghean prepared a fire hazard assessment utilizing the BCTS Strait of Georgia Fire Hazard Assessment Form in May of 2017. He assessed the fuel hazard present on site as follows:

<u>Fuel Loading</u>	<u>Industrial Activity Area Characteristics and Point Rating</u>	<u>Score</u>
Fuel Depth	average greater than 60 cm of dispersed fuel	7
Fuel size	greater than 45% of all fuels <7 cm diameter	7
Horizontal fuel arrangement	greater than 80% fuel coverage of area	7
Vertical fuel arrangement	partially elevated disbursed fuel	3
Vegetation	none	1
Cedar slash	greater than 60% of fuel	4
Risk of ignition	readily accessible to the public	4
Hazard Assessment Subtotal:		33 or High*

[*Low = 5-15 Moderate = 16-26 High = 27-37]

[129] Under "Results", the assessment states that "there is a high fire hazard left on the land base."

[130] Using the Guide for fire hazard assessment, Mr. Boghean generated a score of 100, which he compared to the abatement threshold of 25 based on a severe rating from the BC Fire Risk Map.

[131] Mr. Boghean recommended that further abatement be carried out between the in-block roads and Highway 14. He states that "this is the area of highest

concern and would have the highest impact on surrounding values.” Mr. Boghean stated that the piles in those areas should be reconstructed and burned when conditions permit.

[132] Mr. Boghean also provided a reply expert report in response to Mr. Lougheed’s opinion that the area around Parkinson Creek is not known for historical wildfire activity.

[133] Mr. Boghean prepared a list of 14 wildfires greater than one hectare in size that have occurred since 2012 within 35 kilometres of Parkinson Creek.

[134] In cross-examination Mr. Boghean confirmed that, of the 14 fires compiled in his reply report:

- none of these sites had undergone completed abatement;
- eight of the fires occurred due to escape during prescribed burning;
- four of the fires occurred after piles had been constructed but had not been burned;
- one fire occurred during active logging activity; and
- at least one fire resulted from a trash fire on private land.

[135] In cross-examination, Mr. Boghean also conceded that some of these 14 fires occurred on sites outside of the CWHvm1 BEC zone.

[136] Regarding fire danger days, Mr. Boghean calculated that Parkinson Creek had an average of approximately 30 days of high or extreme fire danger days in the last 10 years, based on weather data obtained from the TS San Juan weather station. He testified that an unnamed weather technician at BC Wildfire Service confirmed that San Juan was representative of Parkinson Creek. However, as this weather technician did not testify at the hearing, the Panel places little weight on this evidence. It is noteworthy, however, that all of the experts who testified, including Mr. Lougheed, used TS San Juan as a representative weather station for Parkinson Creek.

[137] In terms of the fog zone, Mr. Boghean testified that the BC Wildfire Service can exempt coastal areas from campfire bans because, generally, the weather conditions along the coast result in lower fire risk. He also testified that the fog zone boundary begins at Owen Point and carries northward along the west coast of Vancouver Island because, beyond Owen Point, there is generally a lack of infrastructure at risk.

Reconciling the Expert Evidence

[138] The only evidence called by Andersen in support of its assertion that no further abatement is required at Parkinson Creek was the expert evidence of Mr. Lougheed. The remainder of Andersen’s case, at least in terms of the expert evidence, was proffered through the cross-examination of the Respondent’s experts: Mr. Blackwell and Mr. Boghean.

[139] The Panel finds that there are portions of Mr. Lougheed's written report, and his testimony, that undermine the strength of his opinion that no further abatement is required at Parkinson Creek. At least part of Mr. Lougheed's opinion is anchored on the fact that Andersen completed two burn programs on the block. In his view, this alone is evidence of the abatement of any fire hazard, and indicates that conditions at Parkinson Creek do not lend themselves to easy ignition of wildfires.

[140] Mr. Lougheed's written report states that the burn programs were undertaken when weather conditions permitted; however, in cross-examination, he conceded that this was an incorrect statement, and that he now knew that the first burn, performed in April, and the second burn, performed in November, were not ideal for undertaking effective burn programs that would consume the fine materials within the piles.

[141] Mr. Lougheed also said that access to the block had been restricted and that he did not consider public access to be an issue of concern. However, he also conceded that he was able to easily gain access to the block with his vehicle. Mr. Lougheed's report makes no mention of the evidence of campfires having been set within the block.

[142] Accessibility to the public is a cause for concern given the prevalence of human-caused fires. The Panel finds that Mr. Lougheed's lack of concern in this regard undermines his opinion that further abatement is not required.

[143] The fact that Mr. Lougheed had not been to Parkinson Creek to observe the conditions at the time that he prepared his first report also renders his opinions in that report unreliable, since many of his assumptions in that report were incorrect.

[144] One of the most challenging aspects of Mr. Lougheed's evidence concerns the number of burn piles on site and their condition, which again seems to inform his opinion that no further abatement is required. Mr. Lougheed said that he saw 68 piles during his visit and that, of these, 52 of them were completely abated, meaning that they were 85% to 95% burned.

[145] However, Mr. Blackwell testified that he counted 57 piles, of which only 25 had been partially burned.

[146] The Panel notes that, although these experts disagreed somewhat on the number of piles observed (68 piles versus 57 piles), their main disagreement is the extent to which these piles had been subject to burning, and how much of those piles had been consumed (52 versus 25).

[147] According to Mr. McKinnon's evidence, during the first burn program in April, 2015, his abatement plan was to burn a portion of the piles. According to Mr. Hope's evidence, the first burn program was largely unsuccessful. Based upon this unchallenged evidence, there were clearly unburned piles after the first burn: piles that were never designated to be burned, and piles that did not burn. In terms of the second burn program, the burn map prepared by Mr. McKinnon only designated 26 piles for burning. Assuming that there were 68 piles on site as stated by Mr. Lougheed, and only 26 of them were designated for burning, it seems more likely than not that more than half of them were never subject to any burning.

[148] Considering this evidence, it is difficult to reconcile Mr. Lougheed's evidence that he considered that 52 piles had effectively been completely consumed. For that to be the case, all of the 26 piles Mr. McKinnon designated in the second burn, and another 26 piles from the admittedly unsuccessful first burn, were largely consumed. The Panel finds that the totality of the evidence is more supportive of Mr. Blackwell's observations that there were only 25 partially burned piles, leaving some 32 to 40 piles unburned piles depending on whether there were a total of 57 piles or 68 piles. The Panel finds that Mr. Blackwell's assessment of the number of unburned piles to be more credible.

[149] The quantity of piles and their condition puts into question the soundness of Mr. Lougheed's opinion that no further abatement was required. In the Panel's view, his opinion is significantly undermined if there were more than 30 unburned piles remaining.

[150] In reaching his conclusion that no further abatement was required, Mr. Lougheed relied on:

- (a) the belief that two burn programs had largely consumed all of the fine material on site;
- (b) there were only four piles remaining that could potentially be subject to a further burn; and
- (c) any fuel hazard remaining on site that was not consumed previously was due to wet conditions.

[151] These three assumptions are undermined if there were some 30 unburned piles.

[152] The last issue of concern with Mr. Lougheed's opinion is his decision to treat Parkinson Creek as a low risk of fire, notwithstanding that it is found in the severe fire rating category according to the BC Fire Risk Map.

[153] Mr. Lougheed testified that he is generally familiar with weather conditions on the west coast of Vancouver Island, and that he considered Parkinson Creek to be in an area which experienced wet, humid and foggy conditions for much of the year. He agreed that fire danger days are an important consideration when determining fire risk, and that Parkinson Creek will dry out during periods associated with what are considered extreme or high fire danger weather conditions. While Mr. Lougheed reviewed the weather data from the TS San Juan weather station, and concluded that Parkinson Creek likely experienced an average of only 7.8 days per year in high or severe fire danger weather, this was a mathematical error. In cross-examination, he conceded that Mr. Blackwell's figure of high fire danger days was correct at 26 days.

[154] It is apparent that Mr. Lougheed's opinion that Parkinson Creek should be classified as low fire risk was based on: (a) the low number of fire days, and (b) the lack of fine material remaining on site after what he believed were two successful burn programs that had consumed the majority of fine material on site. However, as noted above, both of these factors were incorrect. The better evidence is that there were over 30 unburned piles on site containing fine material that had never been subject to burning in either of the two burn programs, and that the number of

fire danger days was approximately four times greater than Mr. Lougheed had assumed. Accordingly, the Panel finds that Mr. Lougheed's decision to treat Parkinson Creek as a low risk of fire is fundamentally flawed.

[155] As noted previously in these Reasons, Mr. Blackwell opined that there was a sufficient volume of dispersed slash across the site, and a sufficient number of unburned piles, to warrant further abatement. In his opinion, further abatement was necessary to address a risk of ignition in a block close to the urban interface.

[156] Mr. Blackwell was subjected to a vigorous cross-examination. Mr. Blackwell, like Mr. Lougheed, agreed that fire danger days are a significant consideration, and that the classification of those days is dependent upon weather conditions. Mr. Blackwell agreed that local weather at Parkinson Creek could vary from that which was recorded at the TS San Juan weather station. He agreed that fog was unlikely to go 10 kilometres inland to the San Juan weather station. However, fire danger readings are typically taken at 1:00 pm in the afternoon, and Mr. Blackwell did not consider marine fog, typically present in the morning hours, to be particularly relevant to fire danger.

[157] To illustrate the assertion that the San Juan weather is not representative of the weather experienced in Parkinson Creek, counsel for the Andersen placed three photographs before Mr. Blackwell, purportedly taken at Parkinson Creek on July 27, 2017. These photographs were not placed into evidence by the person who took the photographs, and no testimony was given concerning their origin. The photographs were accompanied by metadata indicating that they were taken at approximately 9:00 am on July 27, 2017, along with GPS data which identified the location of the photographs within Parkinson Creek.

[158] One of the photographs, IMG 0087, shows a small pool of standing water. The second, IMG 0076, shows dispersed slash and, notably, the presence of overcast skies and some fog.

[159] The third and fourth photographs show some fog, but also the presence of sedge grass. Mr. Blackwell indicated that sedge grass typically grows in wetter soils. He said that he had not observed any sedge grass during his site visit in December, 2017. He was reluctant to accept that the photographs were taken at Parkinson Creek.

[160] When shown the metadata, Mr. Blackwell eventually accepted that the photographs could have been taken at Parkinson Creek, and accordingly, they were received into evidence and marked as Exhibits.

[161] The manner in which these photographs were tendered to Mr. Blackwell, without having heard from the individual who took the photographs, makes it challenging to afford them much weight. The Panel notes that neither Mr. Lougheed nor Mr. Blackwell, who both walked Parkinson Creek extensively in December, 2017 and March, 2018, respectively, observed any sedge grass. Had they done so, the Panel would have expected it to be mentioned in their reports, or their testimony. Neither said that they saw any. The Panel concludes that the photograph that showed the growth of sedge grass, purportedly present in July, 2017, must only be indicative of this grass growing in a very small isolated location within the block.

[162] Mr. Blackwell was also shown the weather data from the TS San Juan weather station for July 27, 2017, which showed a temperature of 16.2° Celsius, humidity of 73%, and a fire danger class rating of 4.

[163] While the photographs purportedly taken on July 27, 2017 at Parkinson Creek reveal nothing about the temperature or humidity on that day, Mr. Blackwell agreed in cross-examination that, at the time the photograph was taken, and given the presence of fog on the site, he considered there to be a low probability of ignition. He agreed that the danger class rating of 4 recorded at TS San Juan on July 27, 2017, was not representative of the conditions at Parkinson Creek at that time the photo was taken. However, Mr. Blackwell also said that the danger class rating is measured at 1:00 pm each day. No evidence was led as to the conditions at Parkinson Creek at 1:00 pm on July 27, 2017. In his earlier testimony, Mr. Blackwell stated that marine fog typically dissipates during the morning and that fine slash debris can quickly dry out as temperatures rise.

[164] No other evidence of the weather at Parkinson Creek was placed into evidence beyond the July 27, 2017 photos.

[165] Accordingly, the Panel is left with the evidence of only one single day at Parkinson Creek to compare to the average danger class days recorded at TS San Juan and utilized by Mr. Blackwell, and for that matter Mr. Lougheed, in the preparation of their expert opinions. Andersen submits that the evidence from July 27, 2017 is sufficient to call into question the reliability of all of the TS San Juan weather information for depicting the conditions likely experienced at Parkinson Creek.

[166] It is noteworthy that, despite this vigorous cross-examination, Mr. Blackwell never resiled from his opinion that further abatement is required at Parkinson Creek. In the Panel's view it is significant that, even if one were to place Parkinson Creek into any of the categories on the Fire Risk Map other than low, every metric used to assess that risk would still result in a numerical value well in excess of the threshold limit. Accordingly, even if the Panel were to accept that the San Juan weather station is not an exact measure of weather conditions at Parkinson Creek, the limited weather data does not rise the level sufficient to conclude that the risk of fire at Parkinson Creek is low. Andersen has failed to submit sufficient evidence in that regard.

[167] The Panel finds Mr. Blackwell to be an exceedingly credible expert witness. His candor throughout both his direct and cross-examination showed that he gave his evidence in a fair and impartial manner. His testimony conveyed a significant depth of knowledge, gained both through study and practical experience, in the assessment of wildfire ignition potential, and in the assessment and abatement of fire hazards. The Panel prefers his evidence over that of Mr. Lougheed.

[168] The Panel accepts Mr. Blackwell's opinion concerning the appropriate use to be made of the Fire Risk Map, and that areas adjacent to urban areas are shown as severe risk because of the consequences to human safety and infrastructure that would result from a wildfire.

[169] The Panel accepts his opinion that local conditions, such as the presence of fog, precipitation, or low temperatures, can reduce the risk of ignition. However, it

is telling that, in Mr. Blackwell's opinion, the climatic conditions at Parkinson Creek did not cause him to conclude that there was a low fire risk during the months of July and August when fire danger days can be expected to occur.

[170] The Panel accepts Mr. Blackwell's opinion that, based upon his observations at Parkinson Creek in December, 2017, there was an unacceptable number of unburned slash piles and significant areas of dispersed Cedar slash, particularly along the in-block roads. The Panel accepts his opinion that the evidence of human activity within the block adds to the concern over the risk of ignition, and the corresponding risk to adjacent interface values.

[171] The Panel accepts Mr. Blackwell's opinion that the S-3 Cedar slash present after harvesting activities presents a heightened risk of ignition, and that this risk will persist for several years while this fine slash material slowly degrades. The Panel also accepts his opinion that a wildfire, if one were to ignite in Parkinson Creek, would spread quickly and present increased challenges for fire suppression efforts.

[172] The Panel is mindful of Andersen's submission that fuel hazards are but one component of a fire hazard assessment, and that consideration must be given to the risk of ignition and the values at risk. The Panel finds that Mr. Blackwell explored those considerations in his hazard assessment.

[173] Accordingly, Mr. Blackwell's opinion is sufficient evidence on which to conclude that, following harvesting activities at Parkinson Creek, there was an increased risk of a wildfire starting and an increase in fire behaviour or fire suppression as those terms are used in the *Regulation*. The Panel finds that Mr. McKinnon's recommendation to Andersen that no further abatement was required was incorrect.

[174] In making this finding, the Panel is not endorsing a rigid adherence to the Guide, which was one of the concerns raised by Andersen. Nor are licensees being held to a standard of perfection. The Panel accepts that fire assessment is the practice of forestry professionals. There is significant judgment to be exercised based on the language of the legislation. However, in assessing the evidence, the Panel prefers Mr. Blackwell's opinion that further abatement is required. Mr. Blackwell has not, in the Panel's view, placed any rigid reliance on the Guide or the BCTS Strait of Georgia Fire Hazard Assessment Form. Rather, those assessments lend support to his conclusions about abatement garnered through his observations of Parkinson Creek. The Panel finds that, pursuant to section 7(3) of the *Act*, a fire hazard existed as a result of an industrial activity at Parkinson Creek at the time the Abatement Order was issued by the Fire Centre Manager, and that circumstances remain at Parkinson Creek which constitute a fire hazard on the block. The Panel finds that an abatement order is required to abate the fire hazard.

[175] Andersen and the Government made significant submissions concerning the meeting between Andersen and the BC Wildfire Service at the Errington Fire Station on June 29th. Each attempted to attribute bad faith motives to the other in attending at the meeting, and to lay responsibility for the failure of the parties to reach agreement on the other's unwillingness to compromise. In the Panel's view, nothing turns on the evidence related to this meeting. The parties fundamentally

disagreed, as they have throughout this appeal, on the issue of whether further abatement was required at Parkinson Creek. The evidence of the meeting has not served to inform the Panel's decision concerning the need for abatement. The Abatement Order was delivered to Andersen the following day.

[176] However, the wording of the Abatement Order itself warrants consideration. It states:

The circumstances constituting the fire hazard are as follows:

Unabated post-harvest roadside accumulations (piles) and dispersed fuels fire hazard associated with timber harvesting operations.

[177] The Panel finds the language imprecise and overly broad.

[178] In Mr. Lougheed's opinion, there is no practical benefit to further abatement because, with the passage of time, the risk of fire would continue to lessen and three growing seasons have passed since harvesting operations. Mr. Blackwell was of the view that the decomposition of fine Cedar fuels present at Parkinson Creek could take up to 10 years and, in the meantime, new growth of grass would present a risk of fire spread until such time as the tree canopy shaded out the grass.

[179] As noted above in these Reasons, the Panel prefers the evidence of Mr. Blackwell over that of Mr. Lougheed. Accordingly, the Panel is unable to accept Andersen's position that there is no practical benefit to undertaking further abatement at Parkinson Creek, despite the fact that considerable time has passed since harvesting operations were completed.

[180] If some of the seedlings planted after harvesting are damaged in undertaking further abatement, this is a small price to pay for reducing the risk of wildfire to an acceptable level.

[181] That said, the Abatement Order seems to cover the whole of Parkinson Creek, whereas the evidence before the Panel is that the slash accumulations and unburned piles of concern are principally located between Highway 14 and the in-block roads. An appropriate abatement plan is set out on Figure 9 of Mr. Blackwell's report. There, he calls for burning of the existing piles located along the in-block roads, as well as piling of accumulated slash in certain designated areas into new piles and then subject them to burning. The Abatement Order, to the extent that it envisions additional abatement activity across the whole of Parkinson Creek, should be varied to bring it in line with the recommendations made by Mr. Blackwell. When the additional abatement is performed, it should be subject to a follow-up fire hazard assessment conducted by the BC Wildfire Service.

DECISION

[182] In making this decision, the Panel has fully considered all of the evidence and submissions made, whether or not specifically referred to in this decision.

[183] Pursuant to section 41(1) of the Act, the Abatement Order is sent back to the decision-maker with directions to vary the order in accordance with the Panel's findings in paragraph 181, above.

[184] As the Abatement Order has been subject to an automatic stay, the decision-maker is further directed to establish a new date for compliance with the order.

[185] The appeal is allowed in part.

“Jeffrey A. Hand”

Jeffrey A. Hand, Panel Chair
Forest Appeals Commission

November 27, 2018